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Modeling-based prediction tools for preimplantation genetic testing of mitochondrial DNA diseases: estimating symptomatic thresholds, risk, and chance of success

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Abstract

Purpose Preimplantation genetic testing (PGT) has become a reliable tool for preventing the germline transmission of mitochondrial DNA (mtDNA) variants. However, procedures are not standardized across mtDNA variants. In this study, we aim to estimate symptomatic thresholds, risk, and chance of success for PGT for mtDNA pathogenic variant carriers.

Methods We performed a systematic analysis of heteroplasmy data including 455 individuals from 187 familial pedigrees with the common m.3243A>G, m.8344A>G, or m.8993T>G pathogenic variants. We applied binary logistic regression for estimating symptomatic thresholds of heteroplasmy, simplified Sewell-Wright formula and Kimura equations for predicting the risk of disease transmission, and binomial distribution for predicting minimum oocyte numbers.

Results We estimated the symptomatic thresholds of m.8993T>G and m.8344A>G as 29.86% and 16.15%, respectively. We could not determine a threshold for m.3243A>G. We established models for mothers harboring common and rare mtDNA pathogenic variants to predict the risk of disease transmission and the number of oocytes required to produce an embryo with sufficiently low variant load. In addition, we provide a table allowing the prediction of transmission risk and the minimum required oocytes for PGT patients with different variant levels.

Conclusion We have established models that can determine the symptomatic thresholds of common mtDNA pathogenic variants. We also constructed universal models applicable to nearly all mtDNA pathogenic variants which can predict risk and minimum numbers for PGT patients. These models have advanced our understanding of mtDNA disease pathogenesis and will enable more effective prevention of disease transmission using PGT.

Keywords Mitochondrial diseases · Mitochondrial DNA variants · PGT · Genetic counseling · Heteroplasmy

Introduction

Mitochondrial diseases are among the most common genetic disorders with an estimated incidence of 1:4300 [1]. Some mitochondrial diseases are caused by maternally inherited variants in mitochondrial DNA (mtDNA). Pathogenic mtDNA variants are estimated to be present in 0.5% of the population; and to date, over 300 variants have been reported to result in devastating neurological diseases such as MELAS and Leigh syndrome [2–5]. As mtDNA is maternally inherited, females carrying mtDNA variants have a risk of transmitting pathogenic variants to their offspring. Nowadays, some reproductive methods that reduce the risk of transmitting mtDNA variants have already been developed such as prenatal testing, preimplantation genetic testing (PGT), and mitochondrial donation [6]. It is worth noting that various options will be appropriate for different individuals, and determining which option is the most appropriate is not always straightforward and necessitates contributions from both mitochondrial and fertility specialists.

Pathogenic variants in mtDNA can exist alongside wild-type mtDNA molecules within cells (heteroplasmy), and variant load can be measured as a percentage of total mtDNA. For most mtDNA pathogenic variants, patients develop symptoms when variant load surpasses a specific threshold, and severity of disease correlates with variant load thereafter [7, 8]. There is considerable variation of

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variant load across oocytes and offspring of a woman due to a bottleneck in mtDNA inheritance, making the transmission of mtDNA diseases unpredictable [9–11]. Preimplantation genetic testing (PGT) allows the determination of variant load in early embryos and the selection of optimal embryos for implantation, reducing the risk of disease transmission. However, to date, there are no standardized PGT procedures for reducing the risk of germline transmission of different mtDNA variants. Specifically, only limited criteria have been established for selecting eligible embryos and determining the number of oocytes required to ensure acquisition of a suitable embryo.

Here, we developed integrative models that enabled us to predict the probability of carriers with different mtDNA variants being symptomatic. We also used these models to estimate symptomatic thresholds for the same variants. In addition, we established standardized tools that predict risk of disease transmission and the number of oocytes needed for retrieval in PGT to reduce this risk. These models can be adapted to numerous mtDNA variants.

Materials and methods

Establishment of common mtDNA variant databases (m.3243A>G, m.8344A>G, m.8993T>G)

m.3243A>G, m.8344A>G, m.8993T>G are the three most common pathogenic mtDNA variants and thus were selected for modeling. A mitochondrial pedigree database was established prior to the development of prediction models. Mitochondrial pedigrees were derived from Mitomap (https://www.mitomap.org/MITOMAP) and the Reproductive Medicine Center, the First Affiliated Hospital of Anhui Medical University.

Quantitative analysis of the mtDNA variant load

DNA was extracted using Magnetic Universal Genomic DNA Kit (Tiangen, China). Whole mtDNA was amplified via PCR using REPLI-g Single-Cell Kit (Qiagen, Germany) using primers 2120F GGACACTAGGAAAAA ACCTTGTAGAGAGAG and 2119R AAAGAGCTGTTC CTCTTTGGACTAACA, and isolated via gel extraction. Amplified DNA was sheared to fragments of 200 bp and repaired using a DNA end repairing agent (NGS Fast DNA Library Prep Set for Illumina, Joy Orient, China). Products were blunt-end-ligated to adapters using T4 DNA ligase (NGS Fast DNA Library Prep Set for Illumina, Joy Orient, China) and amplified by four to six rounds of ligation-mediated PCR, followed by magnetic bead purification (NGS Fast DNA Library Prep Set for Illumina, Joy Orient, China). NGS libraries were prepared and sequenced on the Novaseq6000 platform (Illumina, USA), using 150-bp-long paired-end reads. Quality control was performed and low-quality sequences were removed. Sequences of mtDNA were aligned to the Cambridge reference sequence NC_012920 using BWA. Variants were filtered based on genotype quality and read depth, and reliable variants were referenced against the MITOMAP database.

Classification and selection of pedigrees

All pedigrees were divided into 3 categories: familial, de novo, and uninformative. Pedigrees were considered "familial" if family history was positive (based on clinical symptoms and the presence of mtDNA variants in maternal relatives). Pedigrees were considered "de novo" if the family history was negative, and mtDNA variant levels of a proband's mother and all tested maternal relatives were 0%. The remaining pedigrees were considered "uninformative."

Uninformative pedigrees were excluded from the database. Within familial and de novo pedigrees, all reported mtDNA variants were analyzed using the Mitomap database. Their pathogenicity was predicted by MitoTIP (for rRNA/tRNA variants) [12] and HmtVar (for coding and non-coding/control region variants) [13]. Only those with a status of pathogenicity "confirmed" or "reported," and with prediction of "pathogenic" or "probably pathogenic," were included in the analysis. Furthermore, variants without relevant pedigree information, variants not associated with neuromuscular diseases, and cancer-related variants were excluded. Finally, a total of 47 RNA-related variants and 27 protein-related variants were included. Detailed inclusion and exclusion processes were presented in Fig. S1. mtDNA variant levels of muscle samples from symptomatic individuals carrying the above variants were collected. A total of 269 familial and 97 de novo mtDNA variants were used in the analysis.

D'Agostino and Pearson test and QQ plots showed that neither data set was consistent with a normal distribution (Fig. S2A, Table S1). The mtDNA variant levels of patients with clinical symptoms were higher in familial pedigrees compared with de novo pedigrees ($P < 0.05$) (Fig. S2B and S2C). As a result, only familial pedigrees were included in the analysis.

Correction and processing of variant load data

mtDNA variant load of blood and muscle was averaged for each patient before analysis. For m.8344A>G and m.8993T>G, the mtDNA variant levels in blood and muscle are usually similar and are relatively constant over time [14–16]. For m.3243A>G, as the mtDNA variant load in the blood of carriers normally decreases over time, blood data were age corrected (Eq. 1) [14, 17, 18]. To avoid

over-correcting, only those lower than 95% after age correction were included in the analysis. In consideration of the limitations of detection sensitivity, in the case where a mother was an mtDNA variant carrier with clinical symptoms and her offspring had very low heteroplasmy (close to 0%), variant load was also included in the analysis as 1%.

$$\text{Age}-\text{adjusted blood level} = (\text{Blood heteroplasmy})/0.977^{(age+12)} \quad (1)$$

All variant carriers were divided into two groups, "symptomatic" or "asymptomatic," based on the occurrence of disease. Carriers with mild symptom were considered "symptomatic" unless the symptoms were extremely common in the patient's age group (such as dementia in patients over 80).

Predicting the probability of symptomatic disease and estimating symptomatic thresholds

Binary logistic regression was used for predicting the probability of being clinically symptomatic based on variant load (Eq. 2). Mean mtDNA variant load value from blood and muscle was the independent variable (x) with a probability of being symptomatic as the dependent variable (y). The model included parameters β_0 and β_1, controlling position and the shape of the function, respectively (Eq. 2). According to Bayes' theorem, the proportion of symptomatic individuals in pedigrees (prior probability of being symptomatic) was required to determine the probability of a carrier being symptomatic based on variant load. The prior probability of being symptomatic was used for weighting data, which was estimated by the ratio of symptomatic siblings (excluding probands) to total siblings. Data from the probands were excluded to minimize the ascertainment bias [7].

$$Ln\frac{y}{1-y} = \beta_0 + \beta_1 x \quad (2)$$

Using the symptomatic probability prediction model and setting a 95% or higher chance of being asymptomatic as the cut-off point, the corresponding mtDNA variant load was determined as the symptomatic threshold for an mtDNA variant (s).

Predicting the risk of disease transmission

Due to the positive correlation between symptomatic risk and variant load, a distribution model of mtDNA variant load across offspring was first constructed to predict risk. The simplified Sewell-Wright formula (Eqs. 3 and 4) [19] and Kimura equations (Eqs. 5–7) [20] were used to establish the distribution model. The simplified Sewell-Wright formula is a function of four parameters: $p0$ (variant load of the mother), t (number of generations), *Neff* (effective population size), and V (variance of mtDNA variant load across oocytes/offspring). p_0 and V (both range from 0 to 1) were substituted into the formula to calculate b (bottleneck parameter) (Eq. 3). p_0 and b were then substituted into Kimura equations (Eqs. 5–7) and the distribution of mtDNA variant load calculated. Numerical calculation methods were as previously reported Wonnapinij et al. [20, 21].

$$V = p_0(1-p_0)\left[1-e^{-t/N_{\text{eff}}}\right] = p_0(1-p_0)(1-b) \quad (3)$$

$$b = e^{-t/N_{\text{eff}}} \quad (4)$$

$$f(0) = (1-p_0) + \sum_{i=1}^{\infty}(2i+1)p_0(1-p_0)(-1)\ {}^{i}F(1-i, i+2, 2, 1-p_0)b^{i(i+1)/2} \quad (5)$$

$$\varnothing(x) = \sum_{i=1}^{\infty} i(i+1)(2i+1)p_0(1-p_0)F(1-i, i+2, 2, x)\ F(1-i, i+2, 2, p_0)b^{i(i+1)/2} \quad (6)$$

$$f(1) = p_0 + \sum_{i=1}^{\infty}(2i+1)p_0(1-p_0)(-1)^{i}F(1-i, i+2, 2, p_0)b^{i(i+1)/2} \quad (7)$$

Based on the symptomatic threshold s for specific mtDNA variants and the distribution model, the risk of disease transmission (the cumulative probability of offspring carrying variant load between s and 100%) was calculated.

Establishment of a model to predict the minimum oocytes for retrieval

Suitable embryos for transfer were defined as those with a probability higher than 95% of carrying a variant load below s. Based on the risk of disease transmission (p), assuming that X number of oocytes are required to ensure acquisition of at least A suitable embryos, and that k is the proportion of all the oocytes (including mature and immature oocytes) retrieved during the oocyte collection procedure that can develop into a transplantable embryo (the transplantable embryo means it is considered to have the potential to result in a healthy pregnancy and live birth), an oocyte retrieval prediction model was developed using the binomial distribution (Eqs. 8 and 9).

$$\sum_{i=0}^{A-1} C_{kx}^{i} p^{i}(1-p)^{kX-i} < 0.05 \quad (8)$$

In particular, when $A = 1$, the model can be simplified as

$$X > \frac{\log_{1-p}0.05}{k} \quad (9)$$

Statistics

GraphPad Prism 8 (GraphPad Software Inc., La Jolla, CA, USA) and MATLAB R2021b (The MathWorks, Inc., Natick, MA, USA) were used for statistical modeling and plotting. Data consistent with normal distribution are presented as mean ± standard deviation (SD), otherwise as median and interquartile range. Mann–Whitney tests were used to compare heteroplasmy data. All data were calculated with 95% CIs.

Results

Characteristics of the database

We first divided individuals carrying m.8993T>G, m.8344A>G, or m.3243A>G into 3 datasets, each including symptomatic and asymptomatic carriers, and analyzed their heteroplasmy in blood and muscle samples. D'Agostino and Pearson test and QQ plots showed that none of the datasets were consistent with a normal distribution (Fig. 1A and Table S2). The variant load of symptomatic m.8993T>G carriers was higher than that of symptomatic carriers of m.8344A>G and m.3243A>G (median = 89%, IQR = 78–95% vs. median = 80%, IQR = 64–90% and median = 70%, IQR = 55–81%, respectively, $P < 0.001$) (Fig. 1B and 1C), suggesting that the symptomatic threshold of m.8993T>G carriers is higher than that of m.8344A>G and m.3243A>G carriers.

The distribution of m.8993T>G variant load in the asymptomatic individuals was consistent with normal distribution, while the distribution of variant load in the symptomatic individuals was not (Fig. S3A and Table S3). The variant load of symptomatic individuals was higher than that of asymptomatic individuals (median = 89%, IQR= 78–95% vs. median = 50%, IQR=40–77%, $P < 0.001$) (Fig. S3B and S3C). Analysis of m.8344A>G carriers indicated that variant load was not normally distributed in asymptomatic nor symptomatic individuals (Fig. S4A and Table S4). The variant load of symptomatic individuals was higher than that of asymptomatic individuals (median = 80%, IQR = 64–90% vs. median = 28%, IQR = 1–59%, $P < 0.001$) (Fig. S4B and S4C). The distribution of m.3243A>G variant load was also not consistent with normal distribution in asymptomatic nor symptomatic individuals (Fig. S5A and Table S5). The variant load of symptomatic individuals was higher than that of asymptomatic individuals (median = 70%, IQR = 55–81% vs. median = 42%, IQR = 20–63%, $P < 0.001$) (Fig. S5B and S5C).

Predicting the probability of carriers being symptomatic and estimating symptomatic thresholds for common mtDNA variants

After pre-processing, our database included the heteroplasmy data of 455 individuals from 187 familial pedigrees (Tables S6–S8), encompassing 87 datapoints from 42 m.8993T>G pedigrees (35 clinically asymptomatic and 52 symptomatic), 152 from 37 m.8344A>G pedigrees (88 asymptomatic and 64 symptomatic), and 216 from 108 m.3243A>G pedigrees (85 asymptomatic and 131 symptomatic).

Based on our m.8993T>G data, we calculated the prior probability of carriers being symptomatic as 0.56, β_0 as − 5.451 (95%, CI − 5.816 to − 5.097), and β_1 as 8.395 (95%, CI 7.920 to 8.885). Using the equation $Ln\frac{y}{1-y} = -5.451 + 8.395x$, we estimated the symptomatic threshold as 29.86% for m.8993T>G (Fig. 2A). The area under the ROC curve (AUC) of our model was 0.847 (95% CI 0.834 to 0.860), indicating our model was a good fit to the data (Fig. 2B).

For m.8344A>G, we calculated the general prior probability of being symptomatic as 0.37, β_0 as − 3.827 (95%, CI − 4.006 to − 3.654), and β_1 as 5.463 (95%, CI 5.205 to 5.728). The equation $Ln\frac{y}{1-y} = -3.827 + 5.463x$ suggested carriers with a variant load over 16.15% would be symptomatic (Fig. 2C). In ROC analysis, the AUC of our model was 0.867 (95% CI 0.859 to 0.876), suggesting an excellent fit to the data (Fig. 2D).

For m.3243A>G, we calculated the general prior probability of being symptomatic as 0.54, β_0 as − 1.696 (95% CI, − 1.806 to − 1.588), and β_1 as 4.213 (95% CI, 4.027 to 4.402). However, we failed to determine a threshold for m.3243A>G using the equation $Ln\frac{y}{1-y} = -1.696 + 4.213x$. According to our formula, when variant load was extremely low (close to 0%), the probability of being symptomatic was still 15.50% (Fig. 2E). In ROC analysis, the AUC of the model was 0.761 (95%, CI 0.751 to 0.770), suggesting a suboptimal fit to the data (Fig. 2F). In conclusion, our models presented a better fit for m.8993T>G and m.8344A>G and allowed estimation of symptomatic thresholds.

Applying our models to predict the risk of disease transmission and required number of oocytes

To demonstrate the application of our models, we used them to predict the risk of disease transmission and the number of oocytes required for PGT using data published by Chinnery's group [22]. Brown et al. studied the distribution of variant load across 82 individual primary oocytes from a patient carrying m.3243A>G [22]. This

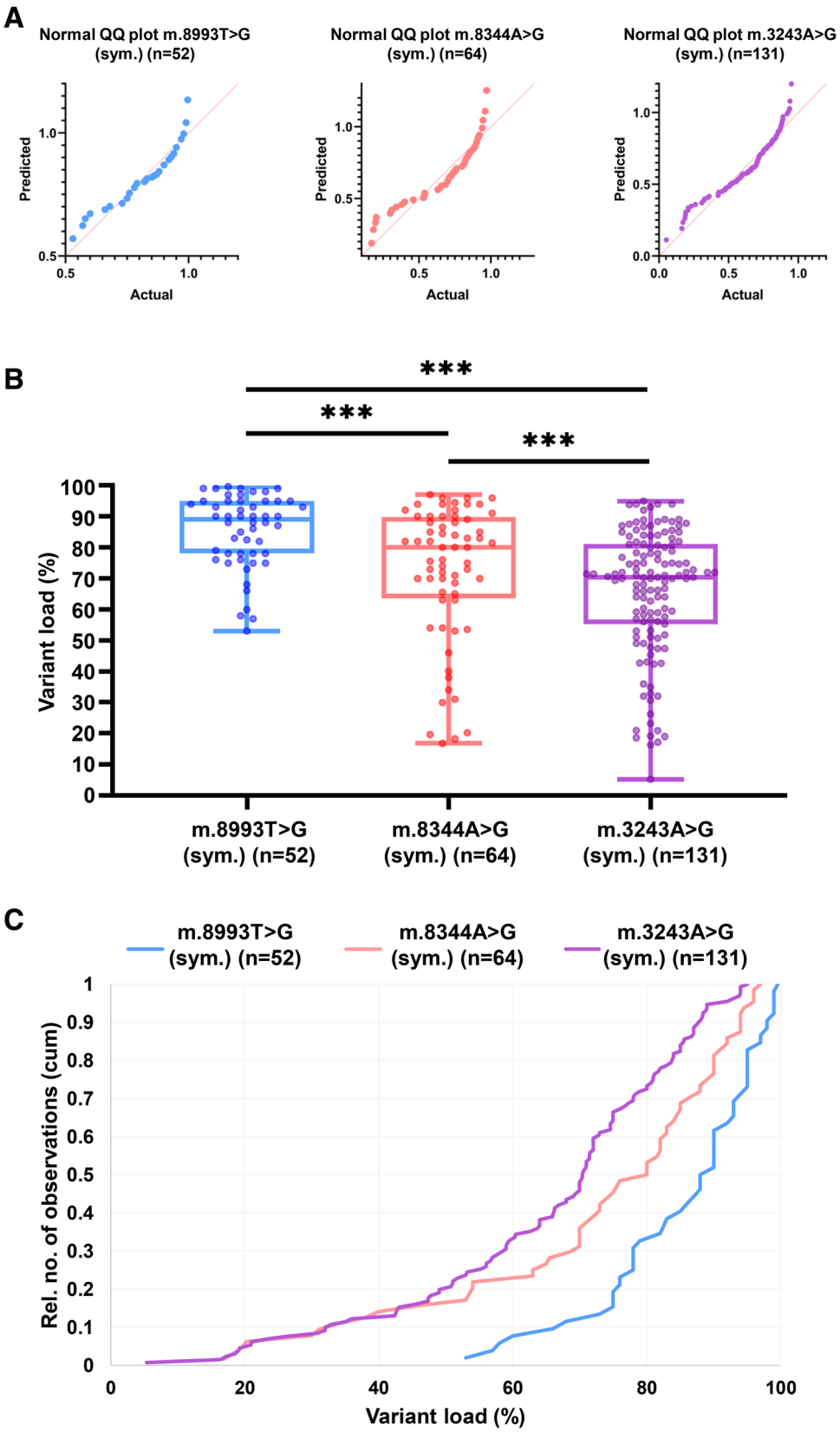


Fig. 1 The distribution of variant load of m.8993T>G, m.8344A>G, and m.3243A>G in the symptomatic individuals (sym.) in familial pedigrees. **A** Quantile-quantile (QQ) plots showing distribution of variant load in symptomatic carriers of m.8993T>G (blue), m.8344A>G (orange), and m.3243A>G (purple). **B** The distribution of variant load of symptomatic individuals for each variant (minimum, 25% quartile, median, 75% quartile, maximum). **C** Cumulative (cum) number of symptomatic individuals relative to all carriers for each variant with variant load equal to or lower than a certain level. *** $P < 0.001$

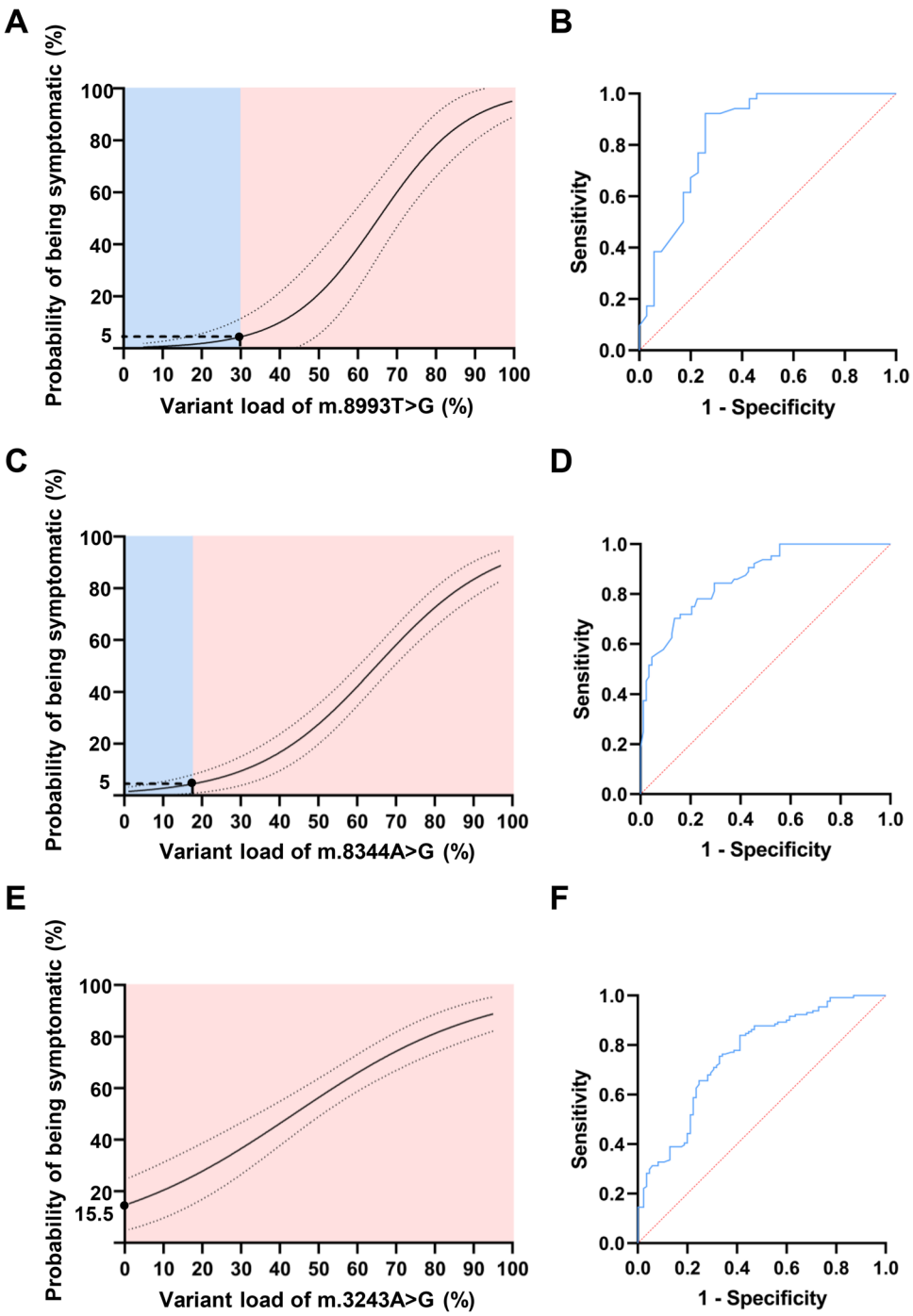


Fig. 2 Prediction model of the symptomatic threshold for common mtDNA variants. **A** Estimated probability of being symptomatic (95% CIs) for specific variant load of m.8993T>G. Carriers with variant load under 29.86% are predicted to be clinically symptomatic by our model (dashed line). **B** The ROC curve of the m.8993T>G threshold prediction model, with an AUC of 0.847. **C** Estimated probability of being symptomatic (95% CIs) for specific variant load of m.8344A>G. Carriers with variant load under 16.15% are predicted to be clinically symptomatic by our model (dashed line). **D** The ROC curve of the m.8344A>G threshold prediction model, with an AUC of 0.867. **E** Estimated probability of being symptomatic (95% CIs) for specific variant load of m.3243A>G. Carriers with variant load of 0% were predicted to have a probability of 15.5% of being symptomatic. **F** The ROC curve of the m.3243A>G threshold prediction model, with an AUC of 0.761. **A**, **C**, **E** *x*-axis represents the variant load of samples and the *y*-axis the probability of an individual being symptomatic. Outer lines represent CIs. The *x*-axis and the *y*-axis are the false positive rate and true positive rate of the prediction model, respectively

dataset was previously used for verifying the Kimura distribution in human oocytes [20]. Our prediction model presented an excellent fit to the data (Eqs. 3–7). The symptomatic threshold of m.3243A>G is generally considered between 15 and 30% [23]. As we were unable to determine a reliable threshold of m.3243A>G using our data, we set an empirical threshold of 15%. In this instance we determined p_0 to be 12.64%, the mean heteroplasmy of the patient's oocytes. This value is more reliable than using the variant load of patients' blood or muscle [19], but the variant load of oocytes is difficult to measure in normal circumstances because the oocytes need to be retrieved surgically. We substituted these parameters into our distribution model (Eqs. 3–7) and calculated the patients' cumulative probability of having a symptomatic child (p) as 65.53% (Fig. 3).

According to data we collected from our intracytoplasmic sperm injection and PGT patients, the proportion of

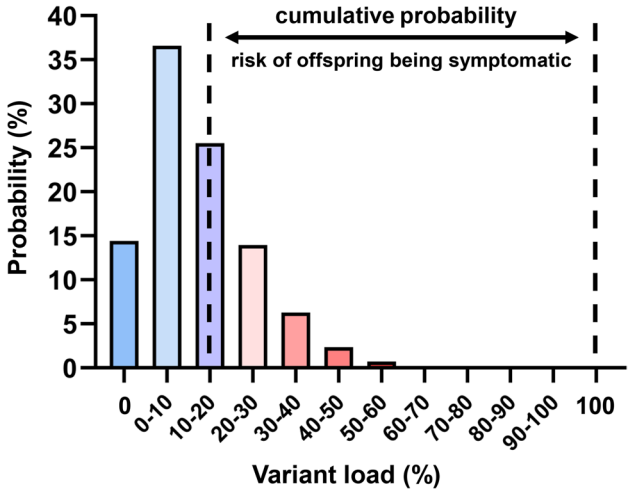


Fig. 3 The probability (*y*-axis) of a child having certain levels of m.3243A>G (*x*-axis) according to our prediction model. The left dashed line represents the symptomatic threshold of m.3243A>G (15%). The right dashed line represents offspring with 100% variant load (fixed). The cumulative probability between the two dashed lines is the risk of offspring being symptomatic

all the oocytes retrieved during the oocyte collection procedure that can develop into a transplantable embryo (k) is 49.3% (Fig. S6). We substituted k and p into our model of oocyte retrieval and found that to obtain at least one embryo with variant load below the symptomatic threshold, the number of oocytes required for retrieval must be more than 14.28 (Eqs. 8 and 9). This indicates that a patient carrying the m.3243A>G variant at an average of 12.64% in oocytes needs to provide minimally 15 oocytes to obtain an embryo with sufficiently low variant load for implantation. Our model can similarly be applied using patient's blood or muscle variant load.

Universal models to predict transmission risk and minimum oocyte number

As shown above, for common variants, we could take the advantage of deep pedigree data to build precise models for the prediction of symptomatic thresholds, risk of disease transmission, and necessary oocyte retrieval numbers. However, it is unrealistic to establish individual models for every rare variant with limited pedigree data. We therefore programmed two universal models for predicting the risk of disease transmission and requisite oocyte retrieval numbers that can be applied to any heteroplasmic mtDNA pathogenic variants, based on variant load in muscles and/or blood of patients. We set the universal threshold value s as 18% and the bottleneck parameter b as 0.66, as these values fit multiple datasets of different mtDNA variants and are applicable to most variants [21, 23, 24].

We modeled the data of three patients from our clinic with rare pathogenic mtDNA variants (P1: m.9185T>G; P2: m.3697G>A; P3: m.10191A>G) (detailed information in Fig. 4A). We predicted the probability of patients' offspring inheriting different variant levels, risk of disease transmission without intervention, and the minimum oocytes needed. We determined (P1) probability of disease transmission is 61.08%, and the minimum number of oocytes necessary for PGT is 13; (P2) probability of disease transmission is 84.64%, and the minimum number of oocytes necessary for PGT is 37; (P3) probability of disease transmission is 28.55%, and the minimum number of oocytes necessary for PGT is 5 (Fig. 4B–4D).

In addition, we calculated the risk of disease transmission and the minimum number of oocytes required for patients carrying any mtDNA variant across a wide range of variant levels (1–60%, Fig. 4E). For patients with over 60% variant load, data were not presented due to high likelihood of high variant load and thus low chance of obtaining a suitable embryo.

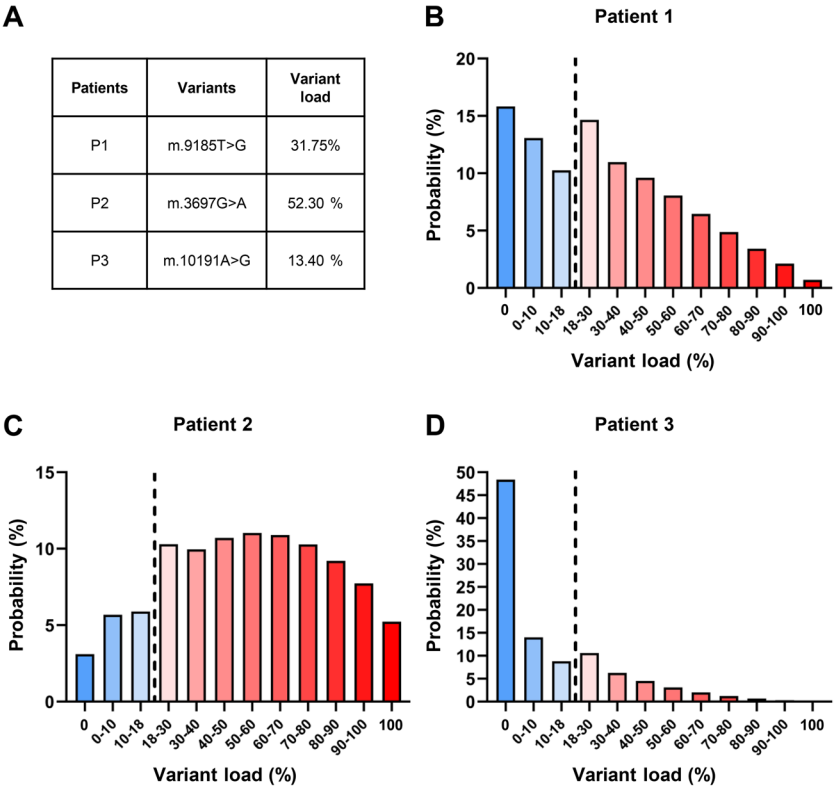
Discussion

For women carrying heteroplasmic mtDNA variants, it is difficult to predict the risk of having a symptomatic child due to a bottleneck in mtDNA inheritance and ambiguous symptomatic thresholds of variant load [11]. PGT enables the measurement of mtDNA variant load in early embryos and reduces the risk of disease transmission. In this study, we have established integrative models that allow us to predict the probability of being clinically symptomatic based on variant load in blood/muscle, and estimated the symptomatic thresholds for m.8993T>G and m.8344A>G. In addition, we have constructed universal models that allow us to predict the risk of disease transmission and the number of oocytes requisite for PGT. These models can be applied to generate standardized PGT guidelines.

Identifying oocytes with the lowest variant load is crucial to successfully conduct embryo transfer during PGT. Mitochondrial and fertility specialists must strike a balance between variant load and the quality of embryos/blastocysts, as well as the likelihood of successful pregnancy upon implantation. However, the cut-off point is unclear for most mtDNA variants, potentially increasing difficulties of genetic counseling and risking disease transmission [25]. Using the collective data of 455 individuals from 187 familial pedigrees, we established models for the three most common pathogenic mtDNA variants (m.3243A>G, m.8344A>G, and m.8993T>G) to predict the probability of a carrier showing symptoms based on muscle and/or blood heteroplasmy levels, and estimated the symptomatic thresholds of these variants.

A

Patients	Variants	Variant load
P1	m.9185T>G	31.75%
P2	m.3697G>A	52.30 %
P3	m.10191A>G	13.40 %



E

Variant load	Risk of transmitting disease	Minimum number of oocytes for PGT	Variant load	Risk of transmitting disease	Minimum number of oocytes for PGT
1%	2.19%	2	31%	59.95%	12
2%	4.37%	2	32%	61.45%	13
3%	6.56%	3	33%	62.91%	14
4%	8.73%	3	34%	64.35%	14
5%	10.90%	3	35%	65.74%	15
6%	13.06%	3	36%	67.11%	16
7%	15.20%	4	37%	68.43%	17
8%	17.33%	4	38%	69.74%	17
9%	19.45%	4	39%	71.00%	18
10%	21.55%	4	40%	72.23%	19
11%	23.63%	5	41%	73.43%	20
12%	25.70%	5	42%	74.60%	21
13%	27.75%	5	43%	75.73%	22
14%	29.76%	6	44%	76.83%	24
15%	31.76%	6	45%	77.90%	25
16%	33.74%	6	46%	78.93%	26
17%	35.69%	6	47%	79.94%	28
18%	37.70%	7	48%	80.91%	29
19%	39.51%	7	49%	81.85%	31
20%	41.38%	7	50%	82.77%	33
21%	43.22%	8	51%	83.65%	35
22%	45.03%	8	52%	84.50%	37
23%	46.81%	9	53%	85.32%	39
24%	48.57%	9	54%	86.11%	41
25%	50.29%	9	55%	86.87%	44
26%	51.98%	10	56%	87.61%	46
27%	53.64%	10	57%	88.31%	49
28%	55.27%	11	58%	88.99%	53
29%	56.86%	11	59%	89.64%	56
30%	58.42%	12	60%	90.27%	60

Fig. 4 Prediction of risk of disease transmission and minimum number of oocytes required for PGT using our universal prediction model. **A** Rare pathogenic mtDNA variants carried by patients and mean mtDNA variant load in blood and muscle. **B** The probability of a patient 1 having offspring with different variant levels. **C** The probability of a patient 2 having offspring with different variant loads. **D** The probability of a patient 3 having offspring with different variant loads. Dashed lines indicate the universal value of pathogenicity threshold (18%). **E** Assuming that every mtDNA variant follows the same role of transmitting pattern through the bottleneck, for patients with certain variant load (1–60%) of any mtDNA variant, risk of disease transmission, and minimum number of oocytes required for PGT were modeled

For most pathogenic mtDNA variants, variant load in skeletal muscle and blood correlates with the severity of clinical presentations, m.8993T>G being one such example [8, 26, 27]. White et al. indicated that the symptomatic cut-off point of m.8993T>G is 60% [27]; however, this prediction can be further refined. A critical difference between our study and that of White et al. lies in the method of categorizing the clinical condition. White et al**.** categorized symptoms into "mild" and "severe" and did not include the asymptomatic carriers, who would likely carry low variant load. Therefore, the proportion of high variant load carriers in the cohort is higher than those in the wider population, compromising the estimation of the symptomatic threshold. This may result in an overestimation of the threshold. In contrast, we categorized carriers by "symptomatic" and "asymptomatic," which improved the precision of the threshold prediction. Our results further define a symptomatic threshold and advance our understanding of phenotype development.

Reports have shown a correlation between clinical features and the variant load of m.8344A>G, but no stringent symptomatic threshold has been determined [17]. Chinnery et al. examined 55 symptomatic m.8344A>G patients, none of whom carried variant load below 50% [17], suggesting an ambiguous threshold of below 50%. van de Glind et al. reported an m.8344A>G carrier with variant load of 15%. The patients' physical examination and laboratory evaluation were normal and they presented with only muscle weakness, raising questions regarding diagnosis [28]. Here we report a more precise symptomatic threshold for m.8344A>G of 16.15% for the first time.

For m.3243A>G, a threshold could not be estimated. Our data and other studies have shown that the m.3243A>G phenotype is not simply correlated to variant load. Recently, factors affecting the phenotypes of m.3243A>G variant have been reviewed and we concluded that mtDNA copy number, nuclear genetic factors, and sex difference also play roles in determining phenotypes among m.3243A>G patients [29]. It is worth noting that individuals might still show symptoms even if their variant load is extremely low (close to 0%) [8, 30]. For example, Damian et al. reported that a 66-year-old patient whose variant load was 5% showed mild neural deafness [31]. In addition, some symptoms such as diabetes, stroke, and myopathy can also derive from other common age-related diseases and a large number of such pedigrees have been reported [32–34], which may confound analysis of disease phenotype. Currently, the threshold of m.3243A>G in clinical practice is considered between 15 and 30% [23, 35, 36]. Given the limitations of the reasons discussed above, precisely defining this threshold requires further investigation.

It is widely accepted that the segregation of mtDNA across oocytes is mostly determined by random genetic drift [22, 36, 37]. The Kimura distribution model of mtDNA inheritance, which can explain the variation in variant load across offspring, was established based on this theory [20, 24, 37]. Using a simplified Sewell-Wright formula, Kimura distribution, and symptomatic thresholds, a risk of disease transmission prediction model can be calculated using variant load of the mother and the variant variance of her oocytes/offspring. Using binomial distribution, the proportion of in vitro fertilized oocytes that can develop into a blastocyst, and the calculated risk of disease transmission, we constructed a model for predicting the minimum number of oocytes required to that ensure at least one embryo with variant load below the symptomatic threshold can be collected. Importantly, these models are less reliable if the calculation of the key parameter V (variance of mtDNA variant load across oocytes/offspring) is based on less than 20 samples [21], thus are unsuitable for rare variants. We therefore established universal models that can be applied to uncommon pathogenic variants, where only a patient's variant load is needed for predictions. The model is constructed based on the assumption that each mtDNA variant follows the same transmission pattern through the bottleneck. We acknowledge that this is a big assumption. Indeed, different mtDNA variants do exhibit varying genetic bottlenecks and even individuals with the same mtDNA variant may possess varying genetic bottlenecks [15, 19]. In addition, our models do not incorporate age adjustment for k. With age there is a decrease in the proportion of in vitro fertilized oocytes that can develop into suitable embryo for transfer. Our models and the prediction table could serve as an initial reference guideline for providing genetic counseling to women harboring pathogenic mtDNA variants. Importantly, they need to be interpreted with caution. Mitochondrial and fertility experts may consult our guidelines and offer personalized advisory opinions, taking into account the specific characteristics of various mtDNA variants, to ensure accurate and reliable data for a given carrier of mtDNA variants.

In addition, we provide a table for clinicians to conveniently calculate risk of disease transmission and minimum number of oocytes required in such cases. For patients carrying a variant load exceeding 60%, data were excluded owing to the higher probability of harboring high variant load and a consequent lower likelihood of obtaining an appropriate embryo. For those carriers with high risk of transmitting mitochondrial disease, PGT should be assessed carefully. They may face increased time and financial burden, and require multiple PGT cycles to get a suitable embryo for transfer. It also highlights the crucial input of both mitochondrial and fertility experts when selecting an embryo for transfer. For those specialists, it is essential to set realistic expectations for mitochondrial disease patients and furnish them with the necessary information to make informed decisions about the most suitable reproductive option for them. In addition, our results also emphasize the significance of

having early discussions on reproductive options for women carrying mtDNA variants. These constraints including the minimum number of oocytes may render many of these mtDNA variant carriers ineligible for PGT, necessitating consideration of alternative reproductive options such as mitochondrial donation. Mitochondrial donation can replace defective mitochondria (contain mutated mtDNA) with healthy mitochondria (contain normal mtDNA), which has been approved by UK in 2015 and Australia in 2022 [38, 39]. Recently, first baby using mitochondrial donation therapy in the UK has been born, with a safe and ethical manner [40]. Although these are still early stage for mitochondrial donation therapy, a bright future for such applications can be foreseen.

Our models will contribute to genetic counseling of women carrying pathogenic mtDNA variants, and further the development of PGT guidelines for clinicians. Importantly, any clinical judgment must be conducted based on adequate genetic counseling for different types of mtDNA variants and varying loads, where mitochondrial and fertility experts play a key role. At the NHS Newcastle Fertility Centre (UK), Dziadek et al. have established an innovative clinical service, providing a model pathway to ensure the provision of the best care and advice for mitochondrial patients [41]. In the future, we will enroll more mtDNA variant carriers to verify our models with further case studies. Recently, we successfully blocked germline transmission of pathogenic variants using PGT in China and the first baby has been born. Our data have been submitted to a peer reviewed journal and are going through scientific peer review. In addition, we have established the largest mitochondrial patient cohort in China and are trying to construct precise prediction models for different mtDNA variants using machine learning. With the accumulation of data, the improvement of modeling tools and the development of artificial intelligence, prediction models will become more precise and personalized.

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Declarations

Ethics approval The study was approved by the Ethics Committee of the First Affiliated Hospital of Anhui Medical University (PJ2020-08-15).

Consent to participate Informed consent was obtained from all individual participants included in the study.

Competing interests The authors declare no competing interests.

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